

RAHUL SHARMA

AUTOMATION ENGINEER

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PROFESSIONAL SUMMARY

Automation Engineer with 4+ years of experience in Software Quality Assurance, including 1 year in manual testing followed by 3+ years specializing in test automation. Began my career validating enterprise Physical Identity & Access Management (PIAM) solutions in the security domain on the Alert Enterprise Guardian platform, gaining hands-on experience in functional, regression, integration, and database testing across workflows involving Apache NiFi, Apache ZooKeeper, Tomcat, and Microsoft SQL Server.

Transitioned into automation engineering, designing and developing scalable, maintainable test automation frameworks using Selenium and Java and then to Playwright, Python, and Pytest. Experienced in building end-to-end UI, API, and regression automation solutions with a strong focus on framework architecture, code reusability, reliability, and CI/CD integration. Proficient in the Page Object Model (POM) design pattern, fixtures, parameterization, reusable utilities, and configuration-driven, modular framework design.

Skilled in API automation and validation — REST API testing, request/response validation, authentication, and schema validation — using Playwright APIRequestContext, jsonschema, Faker, Requests, and pytest-playwright. Experienced in database validation with SQL, SQLAlchemy, and Microsoft SQL Server, integrating automated suites into Jenkins CI/CD pipelines, and using Git/GitHub, Jira, and Agile/Scrum practices for efficient, collaborative delivery.

PROFESSIONAL EXPERIENCE

Alert Enterprise — Chandigarh, IN

Software Engineer II

Jul 2024 – Present

- Leading the migration of the legacy Selenium suite for Airport Guardian to a **Playwright + Python + Pytest** framework, improving execution speed and reducing flaky test failures.
- Designed a modular automation framework using **Page Object Model (POM)**, pytest fixtures, and parameterization, enabling reusable, maintainable test suites across UI and API layers.
- Built API test coverage using **Requests, jsonschema, and Faker** for request/response and schema validation, incorporating data-driven test cases with dynamically generated test data.
- Integrated automated regression and API test suites into **Jenkins CI/CD** pipelines, enabling continuous testing and rapid feedback on every merge request.
- Implemented **SQLAlchemy**-based database checks to verify backend data integrity as part of end-to-end test scenarios.
- Introduced structured **HTML/Allure** test reporting and logging, improving visibility into test run health for cross-functional stakeholders.
- Mentored QA team members on Python, Pytest, and Playwright fundamentals, accelerating the team's shift to modern automation practices.

Software Engineer I

Jan 2022 – Jun 2024

- Achieved a **98% defect detection rate** on Airport Guardian through meticulous test case design and execution, validating enterprise PIAM workflows involving Apache NiFi, Apache ZooKeeper, Tomcat, and Microsoft SQL Server.
- Streamlined testing processes by collaborating with cross-functional teams, reducing product release cycles.
- Developed **Selenium**-based automation test cases covering critical regression scenarios, cutting manual testing effort and testing cycle duration significantly (**by 100% in a few cases**).
- Partnered directly with customers to gather feedback and incorporate it into the QA process, boosting customer confidence.
- Facilitated training sessions for customer support teams on product QA processes, reducing customer-reported incidents by **30%**.

PROJECTS

Playwright + Python Test Automation Framework — github.com/fastasf48-hash/playwright-pytest-automation-framework

- Built a personal end-to-end test automation framework using Playwright and Python, structured around POM, pytest fixtures, and parameterized test data.

- Implemented API testing utilities, configurable environment handling, and structured logging to demonstrate scalable, CI/CD-ready automation practices.

AI Test Failure Analyzer for Playwright — github.com/fastasf48-hash/AI-Test-Failure-Analyzer-For-Playwright

- Built a Playwright + pytest tool that automatically captures screenshots, video, traces, console/network logs, and stack traces on every test failure, with a free rule-based classifier for instant triage.
- Added an opt-in AI analysis layer (OpenAI/Claude, pluggable provider) that generates a structured, schema-validated root-cause report — confidence, severity, suggested fix, and improved locator.
- Built a Streamlit dashboard for execution summaries, failure trends, and flaky-test tracking, linked to pytest-html/Allure reports, with a GitHub Actions CI pipeline that never spends an API call.

SKILLS

Languages: Python, Java, SQL

Automation Frameworks: Playwright, Selenium WebDriver, Pytest (fixtures, markers, parameterization, plugins), Page Object Model (POM)

API Testing: Playwright APIRequestContext, Requests, jsonschema, Faker, pytest-playwright, REST API validation

CI/CD & Tools: Jenkins, Git, GitHub, JIRA, Test Reporting (HTML/Allure)

Database: SQL (MySQL, MS SQL Server, PostgreSQL), SQLAlchemy

Enterprise Platforms: Alert Enterprise Guardian (PIAM), Apache NiFi, Apache ZooKeeper, Tomcat

Concepts: SDLC, Agile/Scrum, OOP, Data Structures & Algorithms, Data-driven & Keyword-driven Testing, Logging & Debugging

Environment & Config: JSON, YAML, CSV, Virtual Environments (venv), pip

EDUCATION

Chandigarh Engineering College (Punjab) — B.Tech, Computer Science — 8.1 CGPA (2022)

DAV Dharamshala (H.P.) — 10th: 9.6 CGPA • 12th: 81.6% (CBSE Board)

ACHIEVEMENTS

- Awarded Quality Champion as fresher for a **98% defect detection rate** during the short time as a manual tester.
- Outstanding Performer Award consecutively for last 2 years during appraisal.